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HOI: A Python toolbox for high-performance estimation of Higher-Order Interactions from multivariate data

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Summary

`HOI` (Higher-Order Interactions) is a Python toolbox to measure higher-order information theoretic metrics from multivariate data. Higher-order interactions refer to interactions that go beyond pairwise connections between nodes in a network (Battiston et al., 2021; Baudot et al., 2019; Gatica et al., 2021; Herzog et al., 2022; Luppi et al., 2024; Rosas et al., 2019). The `HOI` toolbox provides easy-to-use information theoretical metrics to estimate pairwise and higher-order information from multivariate data. The toolbox contains cutting-edge methods, along with core entropy and mutual information functions, which serve as building blocks for all metrics. In this way, `HOI` is accessible both to scientists with basic Python knowledge using pre-implemented functions and to experts who wish to develop new metrics on top of the core functions. Moreover, the toolbox supports computation on CPUs and GPUs. Finally, `HOI` provides tools for visualizing and presenting results to simplify the interpretation and analysis of the outputs.

Statement of need

Recent research studying higher-order interactions with information theoretic measures provides new angles and valuable insights in different fields, such as neuroscience (Baudot et al., 2019; Combrisson et al., 2024; Gatica et al., 2021; Herzog et al., 2022; Luppi et al., 2022), music (Rosas et al., 2019), economics (Scagliarini et al., 2023) and psychology (Marinazzo et al., 2022). Information theory allows investigating higher-order interactions using a rich set of metrics that provide interpretable values of the statistical interdependency among multivariate data (Barrett, 2015; Mediano et al., 2021; Rosas et al., 2019; Scagliarini et al., 2023; Williams & Beer, 2010, 2010).

Despite the relevance of studying higher-order interactions across various fields, there is currently no toolkit that compiles the latest approaches and offers user-friendly functions for calculating higher-order information metrics. Computing higher-order information presents two main challenges. First, these metrics rely on entropy and mutual information, whose estimation must be adapted to different types of data (Czyż et al., 2024; Madukaife & Phuc, 2024). Second, the computational complexity increases exponentially as the number of variables and interaction orders grows. For example, a dataset with 100 variables, has approximately 1.6e5

possible triplets, 4e6 quadruplets, and 7e7 quintuplets. Therefore, an efficient implementation, scalable on modern hardware is required.

Several toolboxes have implemented a few HOI metrics like `infotopo` (Baudot et al., 2019), `infotheory` (Candadai & Izquierdo, 2020) in C++, `DIT` (James et al., 2018), `IDTxl` (Wollstadt et al., 2019) and `pyphi` (Mayner et al., 2018), in Python. However, `HOI` is the only pure Python toolbox specialized in the study of higher-order interactions offering functions to estimate with an optimal computational cost a wide range of metrics as the O-information (Rosas et al., 2019), the topological information (Baudot et al., 2019) and the redundancy-synergy index (Timme & Lapish, 2018). Moreover, `HOI` allows to handle Gaussian, non-Gaussian, and discrete data using different state-of-the-art estimators (Czyż et al., 2024; Madukaife & Phuc, 2024). `HOI` also distinguishes itself from other toolboxes by leveraging `Jax`, a library optimized for fast and efficient linear algebra operations on both CPU, GPU and TPU. Taken together, `HOI` combines efficient implementations of current methods and is adaptable enough to host future metrics, facilitating comparisons between different approaches and promoting collaboration across various disciplines.

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